

Answer Key

Grade 6 Mathematics • Unit 9: Two-variable Relationships • Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 9-1

Independent and Dependent Variables • Tables of Values

Sheet 1 • Independent and Dependent Variables

- a. minutes biking I, miles traveled D b. total cost D, tickets bought I c. guests I, cups of juice D
- Model: "Her distance from home depends on the time she rides the bus." (time riding = independent, distance from home = dependent)
- Independent: number of dogs walked. Dependent: money earned (it depends on the number of dogs).
- a. weeks (on the x-axis) b. height in inches (on the y-axis) c. After 4 weeks, the sunflower is 20 inches tall.
- The size of a number does not decide the variable. The sandwiches needed depend on the number of people, so the number of people is independent and the number of sandwiches is dependent.

Sheet 2 • Tables of Values

- 8, 10, 12, 15
- \$8, \$11, \$14, \$23
- Multiply the boxes by 8 ($y = 8x$); 4 boxes: 32 markers, 8 boxes: 64 markers
- The last value, 8, is wrong. When $x = 4$, $y = 2 \times 4 + 1 = 9$ (the y values go up by 2 each time).
- \$6, \$10, \$14; \$20 is 8 miles ($2 \times 8 + 4 = 20$)

TEST PRACTICE

- C – 29
- B

LESSON 9-2

From Table to Graph • Analyzing a Graph

Sheet 1 • From Table to Graph

- a. (4, 12) b. (0, 0) c. (6, 30) d. (9, 2)
- \$4, \$8, \$12, \$20; points (1, 4), (2, 8), (3, 12), (5, 20) on a straight line through the origin
- a. (5, 10) b. After 3 hours, Kofi has hiked 6 miles. c. 2 miles per hour
- (1, 6), (3, 18), (4, 24). (0, 0) means 0 hours of renting costs \$0.
- He switched x and y. The x value (hours) comes first: plot (2, 10) and (4, 20).

Sheet 2 • Analyzing a Graph

- a. After 4 minutes, the pool has 60 gallons. b. 15 gallons per minute c. 135 gallons
- a. 7 b. 8 c. 12
- Unit rate: $18 \div 3 = 6$ pages per hour; 7 hours: $6 \times 7 = 42$ pages
- Job A: $24 \div 4 = \$6$ per hour. Job B: $18 \div 6 = \$3$ per hour. Line A is steeper, so Job A pays more each hour.
- No. A proportional graph must pass through (0, 0); this one starts at (0, 10). (0, 10) means she had \$10 saved at the start.

TEST PRACTICE

- C – 7
- B

LESSON 9-3

Equations from Tables • Equations from Situations

Sheet 1 • Equations from Tables

- a. $y = 7x$ b. $y = x + 9$ c. $y = 5x$ d. $y = x - 3$ e. $y = x + 6$ f. $y = 9x$ g. $y = 8x$ h. $y = x - 5$
- $y = 11x$; 66, 99
- Each y is 2 less than x: $y = x - 2$. When $x = 20$, $y = 18$.
- $y = 4x$; 10 tickets cost \$40
- It only works for the first row: $3 \times 4 = 12$, not 8. Each y is 4 more than x, so $y = x + 4$.

Sheet 2 • Equations from Situations

- a. $y = 12x$ b. $y = x + 3$ c. $y = 15x + 20$
- $y = 5x + 4$; 4 games: $5 \times 4 + 4 = \$24$
- $y = 2x + 3$; 7 miles: $2 \times 7 + 3 = \$17$
- \$30, \$38, \$46, \$70; $y = 8x + 30$
- She added the fee to the rate, so every hour costs \$35. The fee is paid once: $y = 10x + 25$.

TEST PRACTICE

- B
- C

LESSON 9-4*Substitute and Solve · Working Backward to a Goal***Sheet 1 · Substitute and Solve**

1. a. 19 b. 72 c. 28 d. 42
2. 3, 5, 7, 11, 13; points (0, 3), (1, 5), (2, 7), (4, 11), (5, 13) on a straight line
3. a. $15 \times 6 + 20 = 110$ gal b. $15 \times 10 + 20 = 170$ gal
4. 3 hours: $7 \times 3 + 5 = \$26$; 8 hours: $7 \times 8 + 5 = \$61$
5. a. Yes: $5 \times 4 + 2 = 22$ b. No: $5 \times 6 + 2 = 32$, not 30
6. He added before multiplying. Multiply first: $4 \times 5 = 20$, then $20 + 3 = 23$. $y = 23$.

Sheet 2 · Working Backward to a Goal

1. a. 9 b. 13 c. 9 d. 8
2. $210 - 45 = 165$; $165 \div 15 = 11$ weeks
3. $150 - 30 = 120$; $120 \div 12 = 10$ weeks. Check: $12 \times 10 + 30 = 150$.
4. a. week 4 b. $33 - 6 = 27$; $27 \div 3 = 9$, so week 9
5. $150 \div 12 = 12.5$. She cannot buy half a box, and 12 boxes (144) is not enough, so she buys 13 boxes.

TEST PRACTICE

6. B – 6 hours
7. C – $x = 13$